Hemant Kumar

AI / LLM Engineer · MCP · RAG · Agentic Systems · Python

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SUMMARY

AI/LLM engineer with **16+ years** (since 2010) of software, automation and infrastructure experience, now building **production AI systems in Python**: a 100+ tool **MCP server**, **local RAG** with semantic search, **agentic and multi-agent workflows**, and **evaluation harnesses** that verify whether an agent used its evidence honestly. Ships LLM features into business workflows (OpenAI and Meta APIs inside ERPNext over 74M records) with data-leakage controls by design: fail-closed retrieval, allow-listed corpora, hashed PII, role-based masking and prompt-level guardrails.

100+
tools in an MCP server for LLM agents

74M
records served to an AI-enabled ERP

11k+
automated tests gating an LLM codebase

6,200+
GitHub contributions since 2011

CORE SKILLS

AI / LLM	LLM application design, AI agents and copilots, multi-agent orchestration (Claude Code, Codex, Gemini), MCP servers (stdio + SSE), RAG, vector embeddings and semantic search, prompt engineering and few-shot design, LLM evals (Prime Intellect verifiers, LLM-as-judge), Anthropic & OpenAI SDKs, OpenAI Vision OCR
Self-hosted AI	Ollama (Qwen, nomic-embed-text), GPU inference with PyTorch and Hugging Face Transformers, NVIDIA Canary-1B ASR, silero-vad, Docker Compose model serving
AI security	Prompt-injection and data-leakage aware design: fail-closed retrieval, allow-listed corpora, no egress for sensitive data, SHA-256-hashed PII, role-scoped PII masking, audit logs, verification gates on model output
Backend & data	Python 3.12, FastAPI, Pydantic v2, SQLAlchemy 2, PostgreSQL, Redis, Elasticsearch 8/9, REST/Graph APIs, webhooks, TypeScript, Node.js, Hono, PHP, Frappe/ERPNext
DevOps & quality	Docker, Ansible, Linux, Nginx/Apache, Cloudflare Workers (D1, KV), GitHub Actions CI/CD, Prometheus, Grafana, pytest (xdist), ruff, pyright, import-linter, Vitest, Playwright
Integrations	Meta Graph API (WhatsApp Cloud API, Marketing API custom audiences, Conversions API), OpenAI, Elasticsearch, Playwright automation

EXPERIENCE

Independent AI & Software Engineer · own products and client engagements — 2010 – Present

Designs, builds and operates AI systems, platforms and infrastructure end to end; main author on the engagements below.

Source-bound AI prediction engine · own product — Mar 2026 – Present

Python 3.12 · FastAPI · PostgreSQL · MCP · Anthropic & OpenAI SDKs · Ollama · Docker

- Built an **MCP server with 100+ tools** (stdio for Claude Code, SSE for web clients), layered from data access through domain reads, orchestration and prediction.
- Built a **local, offline RAG service** exposed over MCP: Ollama embeddings and semantic search over an allow-listed corpus; no query or document leaves the machine and the index **fails closed**.
- Wrote a **retrieval-honesty eval suite** (34 deterministic cases, Prime Intellect verifiers): rejects decoys, requires fail-closed answers when a source is missing, and grants tool-use credit only for trace-grounded evidence.
- Enforced "every claim cites a held source" with CI lint gates; **11k+ tests**, ruff, pyright and import-linter architecture rules across ~4,800 commits.
- Runs a **multi-agent engineering workflow** (Claude Code, Codex, Gemini) with a written orchestration policy, per-agent git worktrees and ADRs.

HariDream · microfinance lending platform · AI & Meta integrations 2022 – Present

Frappe/ERPNext · Python · OpenAI · Meta Graph API · Elasticsearch · Next.js 15 · Docker

- Built a **WhatsApp Business app for ERPNext** on Meta's WhatsApp Cloud API: templates, interactive messages, broadcasts, webhooks, catalog and flows, with leads created and assigned in the ERP.
- Added **OpenAI auto-replies** grounded in conversation history and the lead profile (Hinglish, EMI and eligibility maths) with guardrails against promises, OTP/password requests and with human hand-off.
- Automated **KYC with OpenAI Vision**: identity documents received on WhatsApp are read, pattern-validated, stored privately and deleted on a retention schedule.
- Built **Meta Marketing API custom-audience sync** from 10M+ bureau records: idempotent delta-sync, SHA-256-hashed identifiers, batching and rate-limit backoff; no raw PII leaves the system. Added server-side **Conversions API** events to the Next.js loan marketplace.
- Built the **loan-management ERP** over **74M Elasticsearch records** (virtual DocTypes, role-scoped access, PII masking, audit logs) and an **MCP server exposing ERP operations to LLM agents**.
- Built ingestion pipelines merging loan, disbursal and credit-bureau feeds into Elasticsearch with enrichment, Playwright automation and zero-data-loss guards.

AVS Holidays · self-hosted speech-analytics platform Jan 2025 – Jun 2025

NVIDIA Canary-1B · silero-vad · PyTorch · Transformers · FastAPI · Redis · Docker Compose (GPU)

- Led inference and AI services: multi-channel call recordings become transcripts, sentiment and toxicity flags, summaries, topics and suggested next steps.
- Built the voice-activity-detection microservice and the FastAPI/Redis job pipeline; deployed the GPU stack with Docker Compose (~400 commits).

Travel booking platform · UK travel agency 2025 – Present

Cloudflare Workers · Hono · D1/Drizzle · KV · Next.js 15 · React 19 · Vitest · Playwright · Lighthouse CI

- Main author of the API, public site and admin: fare search, quote and enquiry flows, staged imports, security headers and edge caching.

Infrastructure automation · multi-client Linux fleet 2022 – Present

Ansible · Jinja · Docker · Nginx/Apache · TLS · PHP-FPM · UFW · Prometheus · Grafana

- Main author of a 2,100+-commit Ansible codebase that provisions and operates staging and production servers, with central monitoring, firewall drift checks and automated deploys.

ERP products · Frappe/ERPNext v16 2026

- Multi-tenant ERP products (staff operations; real-estate builder suite) with custom apps, role-scoped records and Docker deployment.

Earlier work 2010 – 2023

- **Mobile (2019–2023):** Flutter/Dart apps and plugin contributions (SMS, audio streams, maps, HTTP caching), Firebase and Parse back ends.
- **Home automation & IoT (2014–2016):** built a retrofittable wireless home-automation system (Teensy + XBee + Raspberry Pi on openHAB) and an Arduino sensor-network framework; MQTT on ESP8266, GPS tracking.
- **Computer vision (2015–2017):** face recognition and number-plate recognition experiments with OpenCV, OpenFace and TensorFlow.
- **Web (2010–2019):** education information system in PHP/Zend (2012–2015), WordPress, jQuery and Selenium test automation.

OPEN SOURCE

- **combine-with-context** (main author): VS Code and JetBrains extension that exports a project's code context to Markdown for LLM workflows; published on VS Code Marketplace, Open VSX and JetBrains Marketplace.
- **agent-primer**: two merged PRs hardening CodeGraph MCP setup for AI coding agents (2026). **frappe/telephony**: authored a provider-agnostic telephony-layer proposal (2025).
- Merged fixes to flutter/samples and openface; patches to ArduinoJson, XBee and Flutter plugins; openHAB community member since 2015.